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B.Tech., ComputerScienceand Engineering

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Minor Project- WINE QUALITY CLASSIFICATION

AI-Based Wine Quality Classification

Problem

- Wine quality is traditionally assessed through sensory evaluation by trained experts.
- Sensory assessment can be subjective and time-consuming.
- Wine contains multiple chemical properties such as acidity, sugar, pH, sulphates and alcohol that influence its quality.
- Analysing all these parameters manually is difficult.
- A fast and consistent method is required to estimate wine quality from its physicochemical properties.

Proposed Solution

- Develop a Machine Learning-based Wine Quality Classification System.
- The user enters the chemical composition of a wine through a web interface.
- The trained ML model analyses the input parameters.
- The system predicts the wine quality category.
- The website can display the predicted quality, confidence score and important influencing factors.
- The system acts as a decision-support tool and does not replace certified laboratory or sensory testing.

Impact & Benefits

Faster Screening

Converts wine chemical measurements into a rapid predicted quality category.

Consistent Analysis

The trained model applies the same decision process to every sample.

Data-Driven Decision Making

Uses multiple physicochemical properties simultaneously instead of relying on a single parameter.

Easy Accessibility

A web-based interface allows users to interact with the model without programming knowledge.

Quality Monitoring

Can help identify samples that may require further laboratory or sensory evaluation.

Educational Value

Demonstrates a complete machine-learning workflow from data collection to real-world deployment.

OVERALL IMPACT

Traditional Assessment	Manual Analysis	Time Consuming	Subjective
AI-Based System	Automated Analysis	Faster Screening	Data-Driven Prediction

Key Features & Benefits

Feature	What it does	Benifit
Chemical-input form	11 physicochemical inputs with units/tooltips	Easy and structured data entry
AI Quality Predicrtion	Predicts a defined quality class	Fast preliminary screening
Confidence Score	Shows model probability/confidence	Helps users interpret prediction strength
Feature Importance	Highlights influential variables	Improves transparency and learning
Visual Dashboard	Quality distribution + model metrics	Makes results easier to understand
Model comparison	Compare candidate classifiers	Select the best-performing approach
What-if analysis	Try adjusted values and compare output	Explore how inputs affect classification

Technical Approach

Technology Stack

- **Programming** — Python
- **Data Processing** — Pandas, NumPy
- **Machine Learning** — Scikit-learn
- **Visualization** — Matplotlib, Seaborn
- **Web Application** — Streamlit
- **Model Storage** — Joblib / Pickle
- **Dataset** — UCI Wine Quality Dataset

System Architecture

UCI Dataset
↓
Python + Pandas
↓
Data Cleaning & Preprocessing
↓
Exploratory Data Analysis
↓
Machine Learning Models
↓
Model Evaluation
↓
Best Performing Model
↓
Joblib/Pickle
↓
Streamlit Web Application
↓
User Enters Chemical Composition
↓
Wine Quality Prediction

Research and Reference

Research & References

1. UCI Machine Learning Repository
   - Wine Quality Dataset
   - Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009)
   - The dataset contains physicochemical measurements of wine along with quality ratings.
2. Research Paper
   - Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009).
   - “Modeling wine preferences by data mining from physicochemical properties.”
   - Published in Decision Support Systems, 47(4), 547–553.
   - This research demonstrates the use of data-mining techniques to model wine quality using physicochemical properties.
3. Scikit-learn Documentation

   Used for:
   - Classification algorithms
   - Train/test splitting
   - Model evaluation
   - Confusion matrix
   - Precision, Recall and F1-score

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THE END